

Virtual Reality for Health and Wellbeing of People with Disabilities

Sri Kurniawan
University of California, Santa Cruz
USA
skurnia@ucsc.edu

Sarah E. Anderson
The Ohio State University
USA
anderson.3097@buckeyemail.osu.edu

Aviv Elor
Meta Reality Labs
USA
aviv.elor@meta.com

Jared Duval
Northern Arizona University
USA
Jared.Duval@nau.edu

Alan Lee
Mount Saint Mary's University
USA
ALLee@msmu.edu

Hilary Touchett
Michael E. DeBakey VA Medical
Center
USA
hilary.touchett@va.gov

ABSTRACT

Immersive Virtual reality (VR) offers a transformative medium to enhance health and well-being for individuals with situational, temporary, or permanent disabilities. This panel brings together diverse experts to discuss the latest advances and potential of VR across physical and cognitive health domains. We delve into promising applications for disability-specific support, accessible physical and occupational therapies, and assistive technologies, exploring both opportunities and challenges in inclusivity, user interactions and accessibility. The discussion navigates ethical and interaction considerations and paves the way for future research and development, ultimately aiming to invite developers of VR-based interactive systems and experiences for individuals with disabilities to develop VR interactions that are accessible, user-friendly, and respectful for all users, including those with disabilities.

CCS CONCEPTS

• Human-centered computing → Interaction design process and methods.

KEYWORDS

Immersive Virtual Reality, Disability, Inclusion, Accessibility, Serious Games

ACM Reference Format:

Sri Kurniawan, Sarah E. Anderson, Aviv Elor, Jared Duval, Alan Lee, and Hilary Touchett. 2024. Virtual Reality for Health and Wellbeing of People with Disabilities. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3613905.3643975>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3643975>

1 INTRODUCTION

Immersive Virtual Reality (VR) has the potential to revolutionize the way people with situational, temporary or permanent disabilities (from here all of these conditions are termed “disabilities”) experience the world and maintain or improve their health and wellbeing. For example, VR-based exercises and games can significantly improve motor skills, leading to enhanced mobility, strength, and balance [1, 2, 4]. VR can provide safe and engaging environments for post-surgical support, promoting faster recovery and pain management [3, 10]. Immersive VR environments offer a powerful tool in cognitive exercises for individuals with disabilities, fostering improvements in memory, attention, and problem-solving skills [6, 7, 9]. However, accessibility limitations in VR hardware and software can create barriers for individuals with disabilities [5, 8].

This panel will explore the current state of VR research and development in this domain, with a focus on its potential to address physical and cognitive health challenges faced by individuals with disabilities. Specifically, we will discuss topics such as:

- Disability-specific support: How VR-based exercises and games can be used to improve mobility, strength, and balance, as well as aid in stroke rehabilitation.
- Psychosocial support: How VR can be used to provide immersive and engaging environments to promote psychosocial skills and adaptations, especially for neurodivergent individuals.
- Mental health support: How VR can be used to deliver therapeutic interventions for anxiety, depression, and other mental health conditions, providing a safe and controlled environment for exposure therapy and relaxation techniques.
- Inclusivity and accessibility: How VR can be used to create virtual environments that are accessible to people with disabilities, allowing them to participate in activities that may be physically or socially challenging in the real world.

This panel aims to achieve the following goals:

- Highlight the potential of VR for improving health outcomes and quality of life for people with disabilities.
- Showcase innovative VR applications and research across various areas of health and wellbeing of people with a variety of disabilities.

- Discuss the challenges and opportunities for designing, implementing and evaluating inclusive and accessible VR experiences.
- Foster collaboration and knowledge exchange among researchers, developers, and other stakeholders in the intersection of VR and disability fields.

2 PANELISTS

The panelists will include a mix of researchers and practitioners who have been involved in the development and evaluation of VR systems and interventions for people with disabilities.

Dr. Sri Kurniawan (organizer and moderator) is a Professor in the Computational Media Department at University of California Santa Cruz. Her lab, ASSIST (Assistive Sociotechnical Solutions for Individuals with Special needs through Technology). Her lab performs user-centered design and evaluation of interactive media, including VR and games, for people with disabilities. Her funded projects include VR games for socioemotional skill training for youths with cleft, adults with developmental disabilities, and children with Down Syndrome. She has moderated various panels in the past on similar topics.

Dr. Sarah Anderson OTD, OTR/L (panelist) is an occupational therapist at Nationwide Children's Hospital, and a Postdoctoral Fellow in the NeuroTech Institute and the College of Medicine at The Ohio State University. Her broad area of research is the development and implementation of VR to improve health, safety, and function across clinical, home, and community settings. Her primary populations of interest are pediatrics and adults with neurological and neurodevelopmental motor and cognitive impairments (e.g., pediatric TBI, adult stroke, intellectual and developmental disabilities). Currently, her primary research focus is the reduction of pediatric injury risk by innovating and improving the delivery of home safety intervention for adolescents with brain injuries.

Dr. Jared Duval (panelist) is an Assistant professor in Northern Arizona University's Department of Informatics, Computing, and Cyber Systems, where he directs the Playful Health Technology Lab. He has a Research through Design practice at the intersection of Human Computer Interaction, Games/Play, and Health. For example, one of the games he developed, Spellcasters, is a customizable gesture-based immersive VR game for stroke rehabilitation. Duval represents the viewpoint of a VR system maker in this panel.

Dr. Aviv Elor (panelist) is a senior researcher at Meta Reality Labs leading work on VR input device hardware and accessibility. He is an award-winning IEEE and ACM researcher with a prior focus on VR serious games for physical rehabilitation and has also co-founded Immergo Labs, a startup company that provides one of the world's first VR telehealth platforms for physical rehabilitation. Elor represents the viewpoints of an industry researcher who focuses on innovation on this topic.

Dr. Alan Lee, PT, DPT (panelist) is a Professor in Physical Therapy at Mount Saint Mary's University. He maintains clinical practice at Scripps Mercy Hospital, a level I trauma center, in San Diego with board certifications in geriatrics and wound management as well as certified as a wound specialist. He serves as the current director of the technology SIG in the Academy of Leadership and Innovation. Currently, Lee is co-chairing the telehealth clinical

practice guidelines workgroup and a member of the FiRST council in the American Physical Therapy Association. Lee represents the viewpoints of telehealth practitioners in this panel.

Dr. Hilary Touchett, RN, CRRN, CBIS (panelist) is an investigator for the Center for Innovation in Quality, Effectiveness and Safety (IQuEST) and the South-Central Mental Illness Research, Education and Clinical Education (MIRECC) at the Michael E. DeBakey VA Medical Center in Houston, Texas and an Instructor in the Department of Medicine, Health Services Research Section at Baylor College of Medicine. As a certified rehabilitation registered nurse, Touchett's research focuses on expanding access to care using virtual reality to deliver behavioral health interventions Veterans with Spinal Cord Injury and other Disorders. Touchett represents the viewpoints of rehabilitation nursing and SCI.

2.1 Panel Format

The panel will include 1 moderator and 5 panelists in a hybrid format (some panelists will Zoom in). The panel will be designed as a hybrid event, such that we will use Zoom to interact with both the in-person and virtual audience to moderate questions and discussion. The panel's moderator will act as a facilitator who will guide the discussion and ensure that diverse perspectives are heard. A student volunteer is requested to monitor the Zoom panel to make sure that the voices and questions from the remote participants are heard/conveyed.

The panel will begin with the moderator and panelists providing a brief overview of their work and the potential of VR in their respective fields (3 minutes each). This will be followed by a moderated discussion exploring the following key questions (5 minutes discussion for each question). For each topic, the audience is invited to also express their opinions. It is expected that, because the panelists came from different backgrounds (industry practitioners, healthcare researchers, HCI and VR researchers and developers etc.), there will be opposing viewpoints. It is the responsibility of the moderator to ensure that there will be a lively but respectful debate around each of the topics below.

- What are the most promising applications of VR for health and well-being of people with disabilities, supported by existing research findings?
- What are the challenges and barriers to wider adoption of VR in these settings, including technological limitations, accessibility concerns, and ethical considerations?
- How can we ensure that VR technologies are accessible and inclusive for all individuals with disabilities, drawing on best practices and user-centered design principles?
- What are the ethical considerations involved in using VR for health and wellbeing interventions, particularly regarding data privacy, informed consent, and potential risks of misuse?

The panel will continue with an interactive session where the audience can engage with the panelists and discuss their own experiences and questions. At the end of the panel, the moderator will spend 5 minutes to summarize the comments, ideas, and highlights from the panel to conclude the discussion.

The target audience of this panel are researchers, developers, healthcare professionals, people with disabilities, policymakers,

and anyone interested in the potential of VR to improve the lives of individuals with disabilities.

REFERENCES

- [1] Nicholas Aderinto, Gbolahan Olatunji, Muili Opeyemi Abdulbasit, Mariam Edun, Gbolahan Aboderin, and Emmanuel Egbunu. 2023. Exploring the efficacy of virtual reality-based rehabilitation in stroke: a narrative review of current evidence. *Annals of Medicine* 55, 2 (2023), 2285907.
- [2] Saeed Yousefi Babadi and Hassan Daneshmandi. 2021. Effects of virtual reality versus conventional balance training on balance of the elderly. *Experimental gerontology* 153 (2021), 111498.
- [3] Niamh Brady, Joseph G McVeigh, Karen McCreesh, Ebonie Rio, Thomas Dekkers, and Jeremy S Lewis. 2021. Exploring the effectiveness of immersive Virtual Reality interventions in the management of musculoskeletal pain: a state-of-the-art review. *Physical Therapy Reviews* 26, 4 (2021), 262–275.
- [4] Aviv Elor and Joel Ward. 2021. Accessibility needs of extended reality hardware: a mixed academic-industry reflection. *Interactions* 28, 3 (2021), 42–46.
- [5] Kathrin Gerling and Katta Spiel. 2021. A critical examination of virtual reality technology in the context of the minority body. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [6] Seri Maeng, Jin Pyo Hong, Won-Hyoung Kim, Hyeyoung Kim, Seo-Eun Cho, Jae Myeong Kang, Kyoung-Sae Na, Seok-Hee Oh, Jung Woon Park, Jae Nam Bae, et al. 2021. Effects of virtual reality-based cognitive training in the elderly with and without mild cognitive impairment. *Psychiatry investigation* 18, 7 (2021), 619.
- [7] Maria Grazia Maggio, Rosaria De Luca, Alfredo Manuli, Antonio Buda, Marilena Foti Cuzzola, Simona Leonardi, Giangetano D'Aleo, Placido Bramanti, Margherita Russo, and Rocco Salvatore Calabrò. 2022. Do patients with multiple sclerosis benefit from semi-immersive virtual reality? A randomized clinical trial on cognitive and motor outcomes. *Applied Neuropsychology: Adult* 29, 1 (2022), 59–65.
- [8] Martez Mott, Edward Cutrell, Mar Gonzalez Franco, Christian Holz, Eyal Ofek, Richard Stoakley, and Meredith Ringel Morris. 2019. Accessible by design: An opportunity for virtual reality. In *2019 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 451–454.
- [9] Themis Papaioannou, Alexandra Voinescu, Karin Petrini, and Danaë Stanton Fraser. 2022. Efficacy and moderators of virtual reality for cognitive training in people with dementia and mild cognitive impairment: A systematic review and meta-analysis. *Journal of Alzheimer's Disease* Preprint (2022), 1–30.
- [10] Zina Trost, Christopher France, Monima Anam, and Corey Shum. 2021. Virtual reality approaches to pain: toward a state of the science. *Pain* 162, 2 (2021), 325–331.